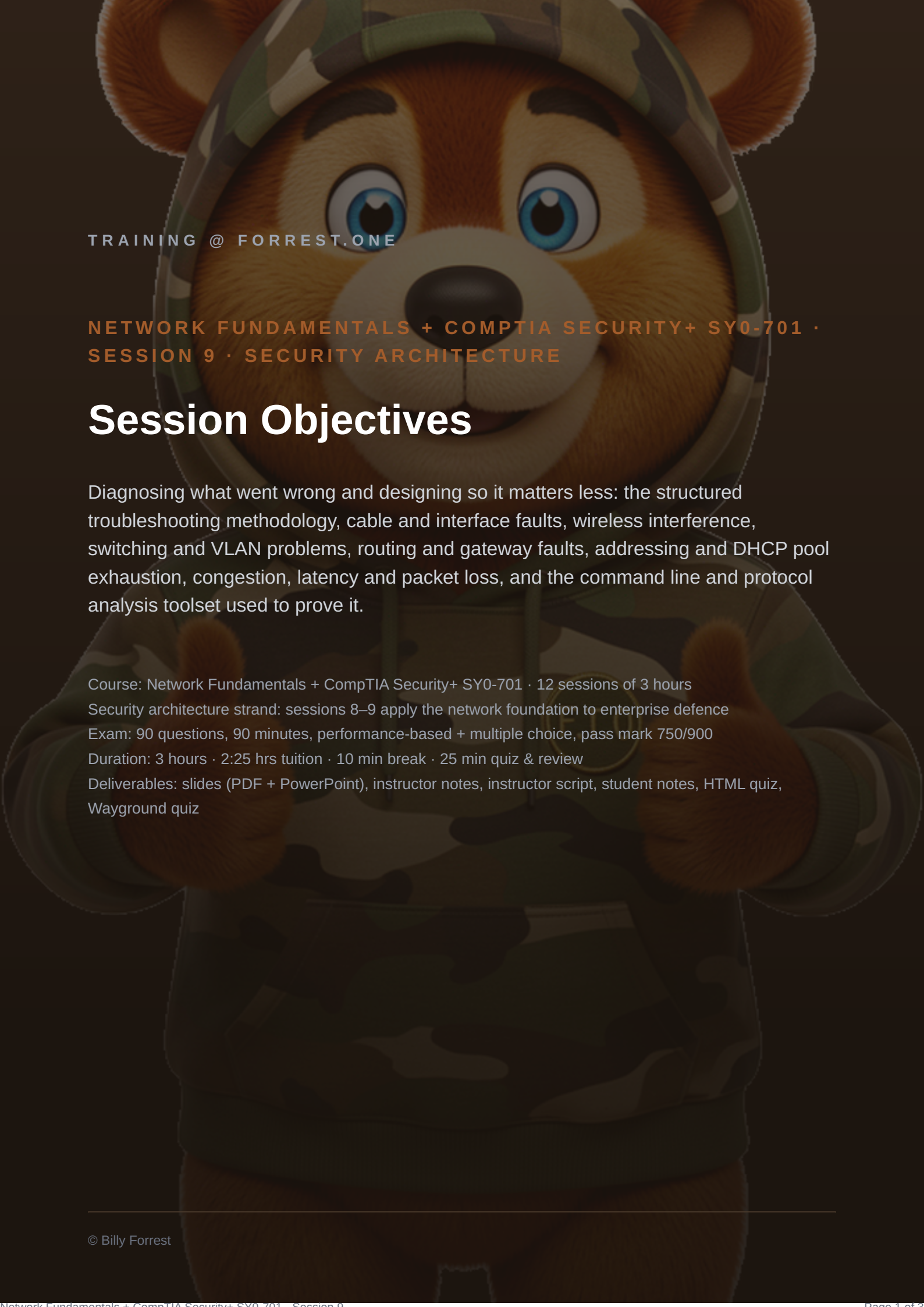




TRAINING @ FORREST.ONE

NETWORK FUNDAMENTALS + COMPTIA SECURITY+ SY0-701 ·
SESSION 9 · SECURITY ARCHITECTURE

Session Objectives

Diagnosing what went wrong and designing so it matters less: the structured troubleshooting methodology, cable and interface faults, wireless interference, switching and VLAN problems, routing and gateway faults, addressing and DHCP pool exhaustion, congestion, latency and packet loss, and the command line and protocol analysis toolset used to prove it.

Course: Network Fundamentals + CompTIA Security+ SY0-701 · 12 sessions of 3 hours

Security architecture strand: sessions 8–9 apply the network foundation to enterprise defence

Exam: 90 questions, 90 minutes, performance-based + multiple choice, pass mark 750/900

Duration: 3 hours · 2:25 hrs tuition · 10 min break · 25 min quiz & review

Deliverables: slides (PDF + PowerPoint), instructor notes, instructor script, student notes, HTML quiz, Wayground quiz

Session Objectives — Endpoint, Application, Cloud & Virtualisation Security

This session extends defence beyond the network boundary to the assets themselves: endpoints, the applications they run, and the cloud and virtualised estate that increasingly hosts them. It covers endpoint protection and management, the secure SDLC, application-layer attacks, and the shared responsibility model.

Learning outcomes

By the end of the 3-hour session, candidates will be able to:

- Describe the endpoint protection stack: AV/EDR, hardening, patching and application control.
- Explain allowlisting versus blocklisting and why allowlisting wins.
- Describe MDM, BYOD models and mobile threats, including remote wipe and containerisation.
- Explain OS security controls: least privilege, secure boot, full disk encryption.
- Describe security in the SDLC: shift left, threat modelling, code review, SAST/DAST.
- Explain DevSecOps and the CI/CD security pipeline.
- Describe SQL injection, XSS, CSRF, buffer overflow, directory traversal and race conditions, and their fixes.
- Explain cloud deployment and service models: IaaS, PaaS, SaaS and public/private/hybrid.
- Explain the shared responsibility model and where the line falls for each service model.
- Describe cloud risks, especially misconfiguration, and the tools that manage them (CASB, posture management).
- Explain virtualisation security: hypervisor attacks, VM escape and container isolation.

Course context

Item	Detail
Course	Network Fundamentals + CompTIA Security+ SY0-701, 12 sessions of 3 hours
Strand	Security architecture (sessions 8–9)
Prerequisites	Sessions 1–3 network foundation; session 8 network security
Assessment	HTML learning quiz (20 questions) + Wayground quiz (15 questions)
Exam link	Domain 3 objectives: endpoint, application, cloud and virtualisation security

Session 9 coverage

ID	Area	What is covered
9A	Endpoint Security	AV/EDR, hardening, app control, MDM, OS security
9B	Application Lifecycle	SDLC, shift left, DevSecOps, SAST/DAST
9C	Application Attacks	SQL injection, XSS, CSRF, buffer overflow, traversal, race conditions
9D	Cloud & Virtualisation	Service models, shared responsibility, misconfiguration, hypervisor security

How this session maps to the course

Session 10 (vulnerability management) will scan the endpoints hardened here. Session 11 (security operations) will investigate the application attacks described here. Every quiz question is tagged with its topic area (e.g. 9C Application Attacks).